

Note that

$$\begin{aligned}s^2 &= M_1^2 + M_2^2 - p_1 p_2 \\ t^2 &= M_1^2 + m_1^2 - p_1 q_1 \\ u^2 &= M_1^2 + m_2^2 - p_1 q_2\end{aligned}$$

Using momentum conservation, we know that

$$p_1 + p_2 = q_1 + q_2$$

and by squaring both sides,

$$-M_1^2 - M_2^2 + 2p_1 p_2 = -m_1^2 - m_2^2 + 2q_1 q_2$$

Now, for any combination we want, we can find it by inner producting one of the variables on momentum conservation condition. For example,

$$q_1(p_1 + p_2) = -m_1^2 + q_1 q_2$$

and substituting for $q_1 q_2$ and $q_1 p_1$ would give $q_1 p_2$.